

Suraj Khonde

Backend Engineer · Node.js · TypeScript · LLM Integration · AWS · Docker

surajrkhonde@gmail.com · +91 7348887772 · Portfolio · GitHub · LinkedIn

SUMMARY

Backend-focused engineer with 3+ years shipping production Node.js / TypeScript services — queues, caching, encryption, Postgres schema design, AWS deploys, and CI/CD. Recent work includes production LLM integration with the Claude API (vision + text) behind confidence gates and guardrails. Currently building and running **Pilooopu**, a live WhatsApp bulk-invite platform at [pilooopu.shop](#), and **Dawa Saathi**, an AI medicine-awareness Telegram bot.

EXPERIENCE

Software Engineer — Advento Technologies, Hyderabad Sep 2025 – Jan 2026

Banking virtual-assistant platform

- Built WebSocket-based bot-to-agent escalation flow handling 1000+ concurrent sessions; handoff latency down ~25%.
- Moved file uploads to presigned S3 URLs so binaries skip the API entirely — server CPU down ~40%, upload time ~20%.
- Rewrote slow report queries with explicit joins and composite indexes; disk I/O down ~35%.
- Containerized the service and set up CircleCI CI/CD with PM2 — repeatable releases with health-checked, near-zero-downtime restarts.

Full Stack Engineer — Technoloader Pvt Ltd, Jaipur May 2024 – Jul 2025

Real-time trading platform

- Streamed filtered Binance market data over Socket.IO with Redis pub/sub fan-out — each session receives only the symbols it subscribed to.
- Restructured the Binance WebSocket consumer into a batched event pipeline; consumer CPU down ~30% at the same per-tick freshness.
- Sustained 10K+ events/min across live updates, webhook ingestion, and reporting under concurrent load.
- Hardened webhook ingestion with idempotency checks and a buffered queue — no dropped events on upstream burst traffic.
- Offloaded reporting and notifications to BullMQ workers with retry/backoff, keeping the real-time price path free of heavy synchronous jobs.

Full Stack Engineer — Avisirah Technologies, Hyderabad Mar 2022 – Feb 2024

Real-time social platform

- Built feed and messaging APIs on Node/Express + MongoDB with Redis as read-cache for hot timelines.
- Implemented JWT access/refresh auth with HttpOnly cookies and RBAC across admin / user / moderator routes.
- Extracted shared middleware (auth, validation, pagination, error formatting) into a reusable layer used across every module.
- Cached high-read endpoints in Redis with TTL invalidation; p50 response time down ~20%.
- Delivered real-time chat, presence, and read receipts over Socket.IO with Redis-backed fan-out across instances.

Equity Research Analyst — non-core roles (2019 – 2022)

TECHNICAL SKILLS

Languages	JavaScript (ES6+), TypeScript
Backend	Node.js, Express, REST, WebSockets / Socket.IO, BullMQ, RabbitMQ, JWT, RBAC
AI / LLM	Claude API (vision + text), prompt design, RAG, confidence gating & guardrails
Databases	PostgreSQL (Drizzle ORM), MongoDB, MySQL, Redis
Infra & DevOps	AWS (EC2, S3), Docker, Nginx, PM2, CircleCI (CI/CD), Swagger / OpenAPI
Tools	Cloudinary, Puppeteer, Razorpay, Pino logging, Firebase (Auth, FCM)
Frontend	React, Next.js, Redux Toolkit, Tailwind CSS

PROJECTS

Pilooopu — Bulk Personalized WhatsApp Invite Platform [pilooopu.shop](#) · [GitHub](#)

Node.js · TypeScript · BullMQ · Redis · PostgreSQL · Drizzle · Puppeteer · WhatsApp Cloud API · Razorpay · Docker · Nginx

- Designed a 3-stage queue pipeline (render → upload → finalize) across 7 BullMQ queues, checkpointed in Redis — a worker crash resumes from the last completed step, not the start.
- Encrypted every guest phone with AES-256-CBC and a unique IV per record; decrypted only inside the send worker, masked as +91*****1234 in every API response and log.
- Split retries into retryable vs non-retryable errors — bad data routes straight to a dead-letter queue instead of burning the 3-attempt budget.
- Reused a single Puppeteer browser across render jobs, producing 400×600 thumbnail and 800×1200 full card from one HTML pass.
- Built a Redis-backed rate limiter with per-endpoint quotas (login 5 / 15 min, signup 3 / hr, event-create 20 / hr) and standard X-RateLimit headers.
- Deployed as two Docker services (API + Worker) behind Nginx; graceful SIGTERM shutdown with 10s force-kill timeout.

Dawa Saathi — AI Medicine Awareness Telegram Bot [t.me/dawasathi_bot](#) · [GitHub](#)

Node.js · TypeScript · Claude Vision + Text · Redis · PostgreSQL · openFDA · Telegram Bot API · Docker

- Built a Telegram bot that reads a medicine-strip photo and returns an elder-friendly drug summary; Claude vision extracts the label at ~80% accuracy.
- Tightened guardrails with a 0.75 confidence gate and ingredient-only prompts that refuse uncertain reads — cutting misleading or wrong drug info ~80%.
- Cached analyzed medicines in Redis → PostgreSQL so repeat scans skip the model — cutting token consumption ~30% and serving answers instantly.
- Dockerized the full app with health checks and graceful shutdown for one-command deploys and fast scaling.

EDUCATION B.E. Mechanical Engineering — VTU, Belagavi (2018)